

Question:
Find the largest possible real part of $((75+117i)z+\frac{96+144i}{z})$ where z is a complex number with $|z|=4$.

Reasoning + Answer:

<think> To maximize the real part of the given expression $((75+117i)z+\frac{96+144i}{z})$, where (z) is a complex number with $(|z|=4)$, we can express (z) as $(4e^{i\theta})$ for some angle (θ) . The expression becomes $((75+117i)4e^{i\theta}+\frac{96+144i}{4e^{i\theta}})=4(75+117i)e^{i\theta}+\frac{24+36i}{e^{i\theta}}$.

Next, we decompose the above into real and imaginary parts. Considering $(e^{i\theta}=\cos\theta+i\sin\theta)$, we have:

$$[4(75\cos\theta+117\sin\theta-i(75\sin\theta+117\cos\theta))+(24\cos\theta-36\sin\theta+i(24\sin\theta+36\cos\theta))]$$

The real part of the entire expression is:

$$[4(75\cos\theta+117\sin\theta)+(24\cos\theta-36\sin\theta)=300\cos\theta+468\sin\theta+24\cos\theta-36\sin\theta=324\cos\theta+432\sin\theta.]$$

To maximize the real part, we use the fact that the real part is maximized when the complex number $(324\cos\theta+432\sin\theta)$ is as large as possible. The expression $(324\cos\theta+432\sin\theta)$ reaches its maximum when it is of the form $(R\cos(\theta-\phi))$ with (R) the amplitude. The amplitude (R) is given by:

$$[R=\sqrt{324^2+432^2}=\sqrt{104976+186624}=\sqrt{291600}=540.]$$

Thus, the maximum value of the real part is 540. We can verify this using trigonometric identity checks if necessary, but since the maximum value occurs at $(324\cos\theta+432\sin\theta=540)$, the maximum real part is achieved.

</think>
<answer>540</answer>

